MEASUREMENT OF ANGLES AND ARCS

12–1 INTRODUCTION

In the first chapter of this book you used a protractor to measure representations of angles. Finally, at this late stage, we prove that the process of angle measurement can be made precise in our geometry. In order to discuss angle measurement, it is convenient first to define the measure of an arc of a circle. This definition is quite like the definition of circumference and really only makes precise the intuitive idea of *length of arc* that you already have. We needed limits to discuss circumference, and of course we also need limits in order to discuss arc length.

Once we know what is meant by arc length, it will be relatively easy to discuss measurement of angles. This discussion will go quite like Chapter 5 on measurement of length of segments.

One reason that a precise description of measurement of angles in degrees is difficult is that describing a unit angle is troublesome. (Is there an angle of one degree?) Furthermore, how do we know that it is possible to divide a degree into 60 congruent angles (minutes) and a minute into successively smaller pieces? This chapter shows how to resolve these difficulties. Incidentally, it should become quite clear that what we choose as the *unit angle* is quite arbitrary, and it is important for you to understand how measurements differ when different units are used. The idea that ties all this together is that of *length of circular arc*.

12–2 CIRCULAR ARCS

We have already described the interior and exterior of an angle (Chapter 3), and we use Definition 3–10 to describe an arc of a circle.

Definition 12–1. If $\angle AOB$ is a central angle of a circle, *the arc* $\widehat{AB}$ *of the circle* subtended by $\angle AOB$ is defined to be the set of points of the circle which are *interior* to $\angle AOB$ or are on the angle. The set of points of the circle which are *exterior* to or on the angle is also called *an arc of the circle* (for example, the arc $\widehat{ABC}$ in Fig. 12–1). If the central angle is a straight angle (like $\angle COB$ in Fig. 12–1), then it has no interior, but the set of points of the circle which are on one side of CB or on CB is called a *semicircular arc* and is said to be subtended by $\angle COB$. Finally, when we speak simply of an *arc of a circle*, we mean an arc subtended by a central angle or the arc exterior to some central angle.

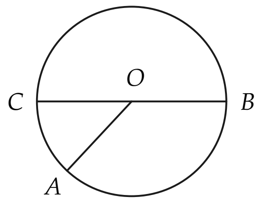

FIGURE 12–1

Exercise Group 12–1

1. In Fig. 12–2, show that if $\angle AOB > \angle COB$, with C in the interior of $\angle AOB$, then every point of the arc $\overset{\frown}{CB}$ is a point of the arc $\overset{\frown}{AB}$.

2. If the arc $\overset{\frown}{AB}$ is subtended by the central angle AOB, define the *midpoint of the arc*, and show that such a point exists.

3. Use the result of Exercise 2 to prove that there are infinitely many points on the arc.

4. In Fig. 12–3, the arc $\overset{\frown}{EF}$ is subtended by $\angle EOF$. Consider the segment EF and any point P on EF. Show that there is, on the ray OP, a point Q, which is on the arc $\overset{\frown}{EF}$. Show that if $P \neq P'$, then $Q \neq Q'$. How many points are there on the arc $\overset{\frown}{EF}$?

5. Referring to Fig. 12–4, suppose that an arc is exterior to a central angle AOB. Show how to define the midpoint of the arc. Why is $\angle AOC \cong \angle BOC$?

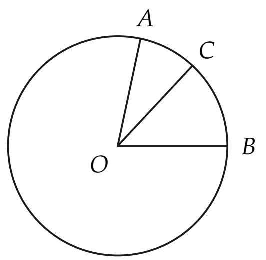

FIGURE 12–2

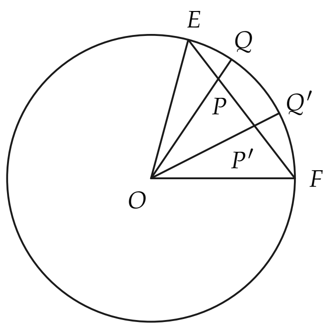

FIGURE 12–3

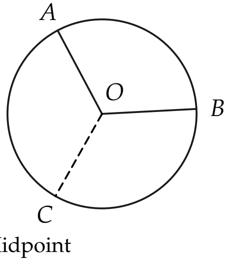

FIGURE 12–4

12–3 ARC LENGTH

In defining the length of a segment, we first had to choose a unit of length. So also in defining arc length, it will be understood that we have already chosen a segment of unit length. We shall also suppose that we have defined the midpoint of an arc as in Exercise Group 12–1.

We describe how to construct a regular inscribed polygonal path in an arc, and indeed, polygonal paths of 2, 4, 8, 16,... sides.

In Fig. 12–5, let AB be a circular arc, and let C be the midpoint of the arc. (See Exercise Group 12–1.) The segments AC and CB form a regular inscribed polygonal path from A to B, with two sides. There are now formed two congruent angles, $\angle AOC$ and $\angle COB$. Each of these angles subtends an arc. Let D and E be the midpoints of these arcs. We can now consider the regular inscribed polygonal path of four sides, $ADCEB$. We then consider each of the arcs $\overset{\frown}{AD}$, $\overset{\frown}{DC}$, $\overset{\frown}{CE}$, and $\overset{\frown}{EB}$ and their midpoints F, G, H, and I. Thus we obtain a regular inscribed path $AFDGCHEIB$ from A to B, having eight sides. Continuing in this manner, for each positive integer we obtain a regular inscribed polygonal path from A to B. That is to say, first we have a path of two sides, next a path of four sides, then a path of eight sides, and so on.

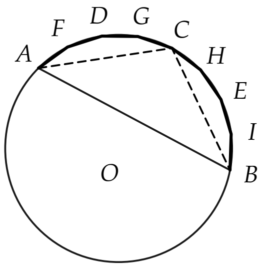

FIGURE 12–5

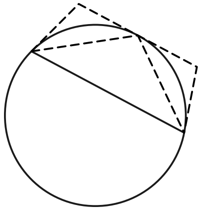

FIGURE 12–6

The following facts are clear from Fig. 12–6, and a complete proof is not difficult to give, although several steps are needed. We shall use these facts without proof. The *length* of the circumscribed path in Fig. 12–6 will be denoted by L. The length of a regular inscribed path of 2, 4, 8, 16,. . . , sides will be denoted by l_n, where n is used to denote the number of segments in the path.

(a) *For every* n, $l_n < L$; that is to say, the length of each regular inscribed path is less than the length of any circumscribed path.

(b) The length of each regular inscribed path of 2, 4, 8, 16,. . . sides is less than the one that follows, that is, $l_n < l_{2n}$.

(c) From (a) and (b) and properties of real numbers, it follows that $\lim\{l_n\}$ exists.

We are now able to define length of arc.

Definition 12–2. Given a circular arc, let l_n be the length of a regular inscribed polygonal path of 2, 4, 8, 16, 32,. . . , sides inscribed in the arc. The *length* l *of the arc* is defined to be the following limit: $l = \lim\{l_n\}$.

Exercise Group 12–2

1. A regular hexagon is inscribed in a circle of radius 1, as in Fig. 12–7. Show that the length of the arc $\widehat{ACB}$ is greater than 2.

2. In a circle of radius 2, what is the length of arc subtended by a right angle? by a straight angle?

3. In Fig. 12–8, the central angles AOB and BOC are congruent, and $\angle AOC$ is a right angle. What is the length of the arc $\widehat{AB}$ if $\overline{AO} = 5$?

*4. Prove that the length of any arc is a positive number which is greater than the length of the line segment determined by the endpoints of the arc.

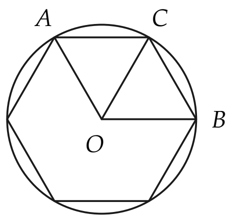

FIGURE 12–7

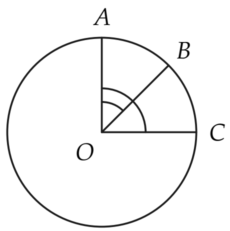

FIGURE 12–8

Theorem 12–1. *If, in a circle, two central angles are congruent, they subtend arcs of equal length.*

Proof. Because the angles are congruent, the inscribed polygons at any stage are also congruent. Thus the lengths of the corresponding inscribed polygonal paths are the same. Hence the arc lengths are the same.

Remark. In the same way it follows that the arcs exterior to the central angles also have the same length.

Having defined arc length, we now would like to prove some more theorems about arc lengths. Indeed, it would be possible to proceed at once to this task, but if we were to do so, our proofs would be quite troublesome. In constructing later proofs it will be most convenient for us to use a theorem (12–2 below) which tells us that in approximating to the circle by inscribed polygons, we need *not* restrict ourselves to regular polygons. The proof of this "helping" theorem is rather hard to make and involves greater knowledge of limits than is presupposed here. Nevertheless, the geometric significance of the theorem should be clear, and it should appeal to your intuition. We shall use this theorem without proof, so that actually we treat it like a postulate.

Theorem 12–2. *Let $\{P_n\}$ be any sequence of polygonal paths inscribed in the arc $\widehat{AB}$ of a circle (Fig. 12–9), and let $\{l_n\}$ be the sequence of lengths of these polygonal paths; let s_n be the length of the longest side of the path P_n. If $\lim\{s_n\} = 0$, then $\lim\{l_n\} =$ length $\widehat{AB} = l$.*

Remark. Evidently, when the length of the longest side of P_n is small, the number of sides, n, must be large. We can now use this theorem to prove some facts about arc length.

Theorem 12–3. *If $\widehat{AB}$ and $\widehat{BC}$ (Fig. 12–10) are adjacent arcs (that is, if they have only point B in common), then length $\widehat{ABC} =$ length $\widehat{AB}+$ length $\widehat{BC}$.*

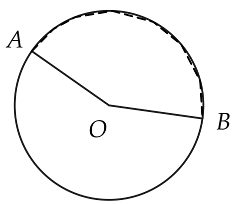

FIGURE 12–9

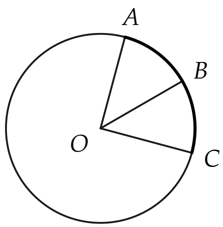

FIGURE 12–10

Proof. The length of $\widehat{ABC}$ is, according to Theorem 12–2, the limit of lengths of inscribed polygonal paths from A to C. Let us always choose these paths so that B is one of the vertices of the path. The path from A to C can then be separated into a path from A to B and a path from B to C. Let l'_n be the length of the path from A to B, and l''_n be the length of the path from B to C. Then, if l_n is the length of the path from A to C, we have $l_n = l'_n + l''_n$.

Now we consider sequences of paths $\{P'_n\}$ and $\{P''_n\}$, with the sequences of lengths of longest sides approaching zero. Thus,

$$\left.\begin{aligned} \lim\{l_n\} &= \text{length } \widehat{ABC}, \\ \lim\{l'_n\} &= \text{length } \widehat{AB}, \\ \lim\{l''_n\} &= \text{length } \widehat{BC}. \end{aligned}\right\} \tag{12–1}$$

But from a theorem on limits, we have

$$\lim\{l'_n + l''_n\} = \lim\{l'_n\} + \lim\{l''_n\}. \tag{12–2}$$

From Eqs. (12–1) and (12–2), we see that

$$\text{length } \widehat{ABC} = \text{length } \widehat{AB} + \text{length } \widehat{BC}. \tag{12–3}$$

This completes the proof. The theorem might be stated briefly as "the length of the 'sum' is equal to the sum of the lengths."

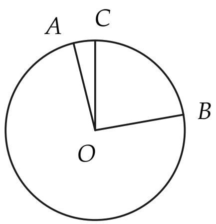

FIGURE 12–11

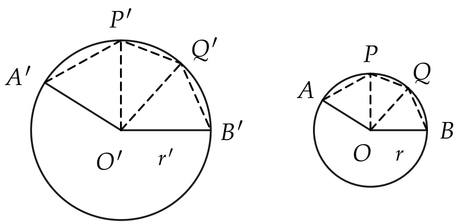

FIGURE 12–12

Theorem 12–4. *If, in a circle, one central angle is greater than another central angle, the greater angle subtends the longer arc.*

Proof. Since congruent angles subtend arcs of equal length, we may suppose that the central angles are as in Fig. 12–11, and that $\angle AOB > \angle COB$. The remainder of the proof (using Theorem 12–3) is left to the student.

Exercise Group 12–3

1. Apply Theorem 12–3 to the arcs subtended by a straight angle and two adjacent right angles.
2. State the form Theorem 12–3 would have for three adjacent arcs; for four adjacent arcs; for n adjacent arcs.
3. Two adjacent arcs in a circle of radius r have lengths $\pi r/3$ and $2\pi r/3$. What angle subtends the two arcs together?
4. Two arcs on a circle of radius r have lengths r and $\pi r/3$. Which is subtended by the greater central angle?

Theorem 12–5 Converse to Theorem 12–4. *If one arc of a circle is longer than a second arc of the circle, the longer arc is subtended by the greater central angle.*

The proof is left to the student.

Theorem 12–6. *Two central angles of a circle are congruent if and only if they subtend arcs of equal length.*

The proof is left to the student.

12–4 ARCS ON DIFFERENT CIRCLES

So far we have compared arcs on the same circle. We now compare the lengths of arcs on different circles. Before studying Theorem 12–7, try to discover the relation between the lengths of two arcs in different circles if they are subtended by congruent angles.

Theorem 12–7. *If $\overset{\frown}{AB}$ and $\overset{\frown}{A'B'}$ are arcs on two circles of radii r and r', respectively, and are subtended by congruent angles, then the arc lengths are proportional to the radii (Fig. 12–12). That is, if $\angle AOB \cong \angle A'O'B'$, then*

$$\frac{\textit{length } \overset{\frown}{AB}}{\textit{length } \overset{\frown}{A'B'}} = \frac{r}{r'}, \qquad \textit{or} \qquad \frac{\textit{length } \overset{\frown}{AB}}{r} = \frac{\textit{length } \overset{\frown}{A'B'}}{r'}.$$

Proof. The proof depends upon three things: (1) the definition of arc length, (2) a theorem on limits, and (3) properties of similar triangles.

Suppose that $APQB$ is a polygonal path inscribed in the arc AB. Then there is a corresponding polygonal path $A'P'Q'B'$ inscribed in the arc $A'B'$ such that corresponding central angles are congruent, that is, $\angle AOP \cong \angle A'O'P'$, $\angle POQ \cong \angle P'O'Q'$, etc. (Why?) Hence, there is a one-to-one correspondence between polygonal paths inscribed in the two arcs. Let L_n be the length of a polygonal path of n sides inscribed in $\overset{\frown}{AB}$, and L'_n be the length of the corresponding polygonal path inscribed in $\overset{\frown}{A'B'}$. Because the corresponding central angles in the two polygons are congruent, for example $\angle AOP \cong \angle A'O'P'$, the triangles formed are similar. (Why?) Since $\overline{AP}/r = \overline{A'P'}/r'$, etc., we have $L_n/r = L'_n/r'$ and $L_n/L'_n = r/r'$.

From the definition of length of arc, length $\overset{\frown}{AB} = \lim\{L_n\}$

and length $\widehat{A'B'} = \lim\{L'_n\}$. From a theorem on limits,

$$\frac{\text{length } \widehat{AB}}{\text{length } \widehat{A'B'}} = \frac{\lim\{L_n\}}{\lim\{L'_n\}} = \lim\left\{\frac{L_n}{L'_n}\right\} = \frac{r}{r'}.$$

This completes the proof.

Exercise Group 12–4

1. A central angle B subtends an arc of length 5 in. on a circle of radius 4 in. A central angle $B' \cong \angle B$ subtends an arc of length 10 in. on another circle. What is the radius of the second circle?
2. A central angle subtends an arc of 2 in. on a circle of radius 5 in. What would be the length of an arc subtended by this same angle on a circle with the same center but with a radius of 2 ft?
3. The radii of two circles have a ratio of 3 to 5. If a central angle subtends an arc of length 2.454 on the first circle, what length arc would a central angle that is congruent to the first angle subtend on the second circle?

12–5 MORE ABOUT ARC LENGTH

In the preceding sections we learned that every arc has associated with it a number called its length. We shall not consider an exhaustive proof of the converse theorem: *For every positive number, less than the circumference, there is an arc having this number as its length.* The complete proof of this theorem requires a rather difficult argument using limits, so we will limit our proof to certain special numbers. It will then be clear that the theorem is true.

In proving the theorem, we confine ourselves to arcs of length not more than one quarter of the circumference, that is to arcs which are subtended by acute angles or right angles.

Theorem 12–8. *Suppose that L is a positive number less than or equal to one fourth of the circumference of a circle of radius r ($L \leqq \pi r/2$).** *Then there is an arc of the circle whose length is L.*

"Proof." If $L = \pi r/2$, then the required arc is subtended by a right angle. If $L = \frac{1}{2} \cdot \pi r/2$, then bisecting a central right angle will give two arcs $\widehat{AC}$ and $\widehat{BC}$, as in Fig. 12–13, of length L. If $L = \frac{3}{4} \cdot \pi r/2$, then the arc AD of Fig. 12–13 has length L, where OD bisects $\angle BOC$. Similarly, if OE bisects $\angle DOC$, then length $\widehat{AE} = \frac{5}{8} \cdot \pi r/2$.

Thus we see that if L is a certain special fractional part of $\pi r/2$, then there is an arc of length L. (Try to describe these fractions.) The fractions for which arcs are easy to construct are 1/2, 1/4, 3/4, 1/8, 3/8, 5/8, 7/8, 1/16,.... In other words, the arc is easy to construct if its length is given simply in terms of halves, quarters, eighths, etc., of a quarter of the circumference.

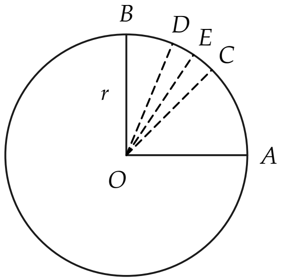

FIGURE 12–13

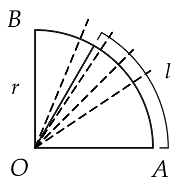

FIGURE 12–14

*Since the circumference of a circle is $2\pi r$, it follows that one fourth of the circumference is $\frac{1}{4} \times 2\pi r$, or $\pi r/2$.

You are familiar with measuring lengths with a ruler whose scale is divided into halves, quarters, eighths, sixteenths, etc. Hence, if the number L is not exactly expressible in halves, quarters, eighths, etc., we can still construct an arc whose length is very close to L.

In Fig. 12–14, L appears to be a little greater than $\frac{5}{8} \cdot \pi r/2$. It is plausible therefore, that for every $L < \pi r/2$, there is exactly one arc AP, as in the figure, with length $\overset{\frown}{AP} = L$.

Exercise Group 12–5

1. In this section we gave a list of fractions whose associated arcs are easy to construct. The list is computed according to a definite rule. Continue the list to 20 more terms.

2. On a circle of radius r, construct arcs of length $7/8 \cdot \pi r/2$; $9/16 \cdot \pi r/2$; $7/4 \cdot \pi r/2$.

3. Suppose that there is an arc of length 3 on a circle of radius 2 in. What part (as a decimal) of $\pi r/2$ is 3? Show how to approximate this decimal by a sum of halves, quarters, eighths, etc. Use the approximation 3.14 for π, and then rework using 3.1416.

4. An arc of length r on a circle of radius r is what part of the circumference? of one quarter of the circumference?

5. On a circle of radius 32 in. is an arc of length 22 in. What part of a right angle does it subtend? Use $\pi = 22/7$. Give a construction approximating this arc.

12–6 MEASUREMENT OF ANGLES

In the previous section we have in effect been measuring angles by measuring arcs. In fact we have expressed the arc length as a fraction of $\pi r/2$ that is as a fraction of one quarter of the circumference. In doing this we have measured the corresponding central angle in terms of the right angle as a unit of angle. In this section we shall define the measure of an angle in terms of other units.

In deciding how to measure angles, we must first define the unit. This involves choosing some angle whose measure is 1. The choice is quite arbitrary, but there are some restrictions based on convenience. If the unit angle is small, the measures of common angles will be large numbers; whereas if the unit angle is large, the measures of common angles will be small numbers. It turns out that the familiar degree is a convenient size unit for ordinary measurement, because the numbers involved are a "good" size. Obviously, other unit angles could be selected. A natural one would be obtained by dividing the circle into 100 arcs of equal length, or perhaps the right angle into 100 congruent angles.

How does it happen that we use that rather strange unit, the degree? We measure angles in degrees because the ancient Sumerians* (2000 B.C. and earlier) did. They divided the circle into 360 "equal parts," perhaps because they thought at one time that the year was 360 days long!

Definition 12–3. On a circle of radius r, an arc of length $1/360 \times 2\pi r$ is subtended by a central angle of one degree (1°).

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*The Sumerians had a highly developed civilization in Mesopotamia, in the valleys of the Tigris and Euphrates rivers, more than 5000 years ago. Roughly contemporary with ancient Egypt, the Sumerian culture discovered much about arithmetic, geometry, and astronomy. We inherit many things from these gifted people. Their method of counting was based on 60 rather than our base of 10. Thus we divide the hour into 60 minutes, and the minute into 60 seconds. They also invented place value in number notation, although they had no "zero." Many of our words go back to Sumerian roots. For more Sumerian history, see *A History of Science*, by George Sarton, Harvard University Press, 1952, or your encyclopedia.

Remark 1. The definition of degree is independent of the radius r, by Theorem 12–7.

Remark 2. Any angle congruent to an angle of 1° is also an angle of 1°, by Theorem 12–6.

Exercise Group 12–6

1. Write out the proof of Remark 1.
2. Write out the proof of Remark 2.
3. Define "minute" and "second."
4. Define a new unit of angle, which we will call the *centile*, by dividing one quarter of the circumference into 100 equal arcs. Using this unit, what is the measure of a 30° angle? a 45° angle? a straight angle?
5. Divide the circumference of a circle into two arcs of equal length. What angle is subtended by each arc? Each arc is what part of the circumference? What is the measure in degrees of the central angle that is subtended by each arc?
6. Divide the circumference of a circle into four arcs of equal length. What kind of an angle is subtended by each of these arcs? Each arc is what part of the circumference? What is the measure in degrees of each central angle?
7. Consider an arc whose length is one sixth of the circumference. What is the measure in degrees of the central angle that is subtended by this arc?
8. Consider an arc whose length is one sixth of the length of a semicircular arc. What is the measure in degrees of the central angle that is subtended by this arc?
9. If a semicircular arc has

length 8 in., and a second arc has length 2 in., what part is the length of the second arc of the first? What is the measure in degrees of the central angle that is subtended by the second arc?

10. On a circle of radius 4, what is the length of an arc subtended by an angle of 1°? of 2°? of 30°?

Definition 12–4. If, on a circle of radius r, a central angle subtends an arc of length L, then the measure of the angle in degrees is $L/\pi r \times 180$.

Remark 1. This definition agrees with Definition 12–3 for angles of 1°.

Remark 2. By Theorem 12–7, the measure of an angle in degrees is independent of the radius.

Remark 3. If M is the measure of the central angle in degrees, we have from Definition 12–4 that $L = M\pi r/180$.

Exercise Group 12–7

1. If $\angle AOB$ and $\angle BOC$ are adjacent angles of 1° each, show that the measure of $\angle AOC$ is 2°.
2. Show that a straight angle has measure 180°.
3. On a circle of radius 6, find the number of degrees in a central angle that subtends an arc of length 3; 6; π; $2\pi/3$.
4. On a circle of radius 9, find the length of an arc subtended by an angle of 30°; 180°; 60°; $(180/\pi)^\circ$.
5. Show that bisecting a right angle gives an angle whose measure is 45°.

6. Prove Remark 3 under Definition 12–4.

12–7 COUNTING DEGREES AND MEASURING ANGLES

Measurement of angles is quite similar to the counting process which led to the idea of length of a segment. In this section we give a description of this process and find that it gives the same measure of an angle as is given by Definition 12–4. Detailed proofs will be omitted. The following theorem shows how counting applies.

Theorem 12–9. *Suppose that $\angle AOB$ is an acute angle, and we consider successive arcs subtending central angles of 1° on the circle from B to A so that the point A is a point of the $(n+1)$st arc (Fig. 12–15). Then the measure of $\angle AOB$ in degrees is either equal to n, or $(n+1)$, or is between n and $(n+1)$; $n \leqq$ measure of $\angle AOB \leqq n+1$.*

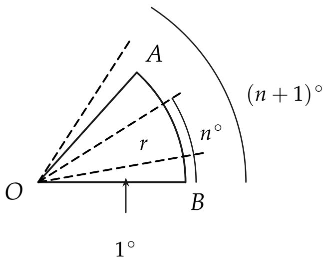

FIGURE 12–15

Proof. The length of the arc of n° is $n \cdot \pi r/180$; the length of the arc of $(n+1)^\circ$ is $[(n+1)\pi r]/180$. The length L of the arc $\widehat{AB}$ is then between these two numbers, by Theorem 12–5.

$$n \cdot \frac{\pi r}{180} \leqq L \leqq \frac{(n+1)\pi r}{180}. \tag{12–4}$$

By Definition 12–4, the measure of $\angle AOB$ in degrees is $L \cdot 180/\pi r$. From Eq. (12–4), we see that

$$n \leqq L \cdot \frac{180}{\pi r} \leqq (n+1),$$

and this is what we wished to prove.

Remark. The measure to the nearest minute could be obtained by counting in a similar way.

Consider an angle $A'O'B'$ (Fig. 12–16) that is congruent to $\angle AOB$ of Theorem 12–9. Consider also successive arcs subtending central angles of 1° on arc $B'A'$, as described for arc BA. If the point A is on the $(n+1)$st arc, then the point A' is on the $(n+1)$st arc. This can be shown using the addition and subtraction of angles postulate and the definition of "less than" and "greater than" for angles. This proves the following theorem:

Theorem 12–10. *If two angles are congruent, they have equal measures.*

The converse of Theorem 12–10 can be proved, using the fact that $\angle O > \angle O'$, or $\angle O \cong \angle O'$, or $\angle O < \angle O'$.

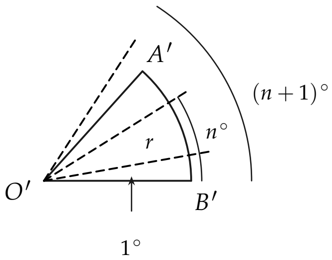

FIGURE 12–16

Exercise Group 12–8

Give your answers in terms of π.

1. If the radius of a circle is 14, what is the length of the arc subtended by an angle of 30°?
2. On a circle of radius 5 in. is an arc of length 6 in. What is the measure in degrees of the central angle subtending the arc? What if the radius is 20 in.? 6 in.?
3. A central angle of a circle subtends an arc of length equal to the radius. How many degrees in the angle?
4. An arc of length 8 in. on a circle of radius 6 in. subtends a central angle of how many degrees? an arc of length 4 in.? an arc of length 6 in.?
5. On a circle of radius π, there is an arc of length $\pi^2/4$. How many degrees in the central angle?
6. An arc of length $\pi r/2$ is subtended on a circle of radius r by what central angle?
7. Find the length of arc on a circle of radius r subtended by an angle of 1°.

We have been measuring arcs by their lengths. It is convenient to measure them by the measure in degrees of the central angles they subtend.

Definition 12–5. The measure in degrees of an arc subtended by a central angle is equal to the measure of the central angle. The measure in degrees of an arc exterior to a central angle is equal to 360 minus the measure of the central angle. (See Fig. 12–17.)

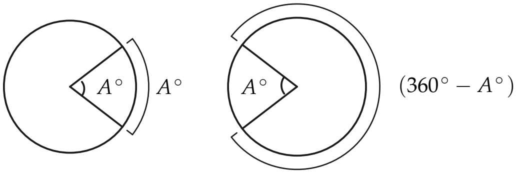

FIGURE 12–17

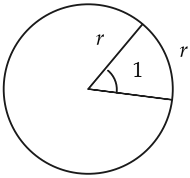

FIGURE 12–18

Exercise Group 12–9

1. An arc of length π in. on a circle of radius 3 in. is of how many degrees?
2. An arc of length 4 in. on a circle of radius 6 in. has how many degrees?
3. On a circle of radius 2 in., how long is an arc of 120°? 240°?
4. How many degrees has an arc of length 15 in. on a circle of radius 3?

12–8 RADIAN MEASURE

Besides the familiar unit of degree for measuring angles, there is another one which is of great importance in advanced mathematics. The reasons for using it cannot be given here, but the definition is quite simple, and you will meet it again if you study trigonometry. The unit is called the *radian*.

Definition 12–6. An angle of one radian is one which subtends an arc of length r on a circle of radius r (Fig. 12–18).

Definition 12–7. The measure of an angle in radians is equal to L/r, where L is the length of the arc subtended on a circle of radius r.

Remark 1. Definition 12–7 agrees with Definition 12–6 for angles of 1 rad.

Remark 2. The measure in radians does not depend on the radius, by Theorem 12–7.

Exercise Group 12–10

1. On a circle of radius 2 is an arc of length 4. How many radians are there in the central angle?
2. How many radians are there in a right angle? in a straight angle?
3. On a circle of radius 2, find the length of arc subtended by a central angle of $30°$. How many radians are there in this angle?
4. How many degrees are there in one radian?
5. How many radians are there in one degree?
6. From Exercises 4 and 5, derive the formulas for changing measure in degrees to measure in radians, and vice versa.
7. What are the measures in radians of the following angles? $30°$; $45°$; $60°$; $90°$; $120°$; $135°$; $150°$; $180°$.
8. The following numbers are measures of angles in radians. What are their measures in degrees? $\pi/4$; $5\pi/4$; 3; 2; 1.5; $\pi/3$; $\pi/6$; $\pi/2$.
9. Which is the larger unit, radian or degree? For which unit will the measure of an angle give the larger number?
10. Consider a central angle of 1 radian on a circle of radius r. What is the length of the arc subtended by this angle? an angle of 2 radians? an angle of 3 radians? an angle of n radians?

Consequence of Definition 12–7. If R is the measure in radians of a central angle in a circle of radius r, then the angle subtends an arc of length $r \cdot R$.

12–9 OTHER UNITS

Units other than degrees and radians could be chosen. One which has some use in artillery is the *mil*. It is approximately the angle which subtends an arc of 1 yd at a distance of 1000 yd. Apart from its military use, it is of little importance.

12–10 INSCRIBED ANGLES

Now that we have a measure for angles, it is easy to prove a theorem about the measure of angles *inscribed* in a circle. We recall the definition.

An angle is said to be *inscribed* in a circle if its vertex and one point on each side of the angle, other than the vertex, lie on the circle. The arc of the circle interior to the angle is said to be subtended by the angle. (See Fig. 12–19.)

Theorem 12–11. *The measure of an inscribed angle in degrees is equal to one half of the degree measure of the arc it subtends.*

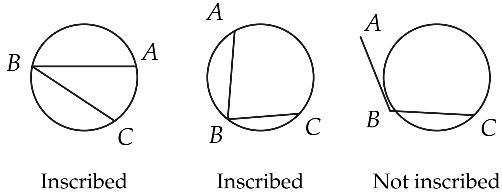

FIGURE 12–19

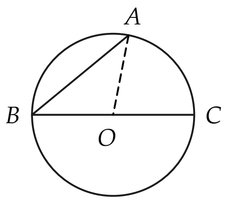

FIGURE 12–20

First, we prove a special case of this theorem, which will enable us to prove the theorem easily.

Lemma 12–11–1. *If $\angle ABC$ is an inscribed angle and BC is a diameter of the circle, then measure $\angle B = \frac{1}{2}$ measure $\widehat{AC}$.* (See Fig. 12–20.)

Proof.

STATEMENTS	REASONS
1. $\angle OAB \cong \angle B$.	Why?
2. Measure $\angle OAB +$ measure $\angle OBA =$ measure $\angle COA$.	Why?
3. 2 (measure $\angle B$) $=$ measure $\angle COA$.	Why?
4. Measure $\angle COA =$ measure $\widehat{AC}$.	Why?
5. Measure $\angle B = \frac{1}{2}$ (measure $\widehat{AC}$).	Why?

Proof of the theorem. There are two cases to consider to complete the proof of the theorem. They are: *Case* 1. The center of the circle is interior to $\angle ABC$ (Fig. 12–21). *Case* 2. The center of the circle is exterior to $\angle ABC$ (Fig. 12–22).

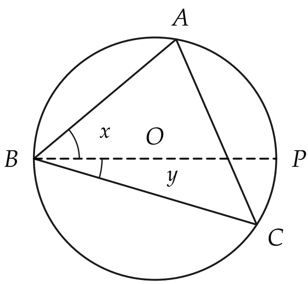

FIGURE 12–21

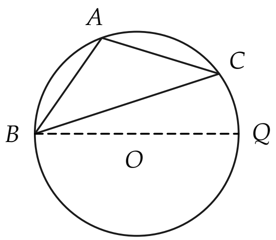

FIGURE 12–22

In Case 1 the diameter through B divides $\widehat{AC}$ into arcs $\widehat{AP}$

and $\widehat{PC}$, and $\angle ABC$ into angles x and y, as shown. By Lemma 12–11–1, measure $\angle x = \frac{1}{2}$ (measure $\widehat{AP}$), and measure $\angle y = \frac{1}{2}$ (measure $\widehat{PC}$). Hence measure $\angle ABC =$ measure $\angle x +$ measure $\angle y = \frac{1}{2}$ (measure $\widehat{AP}$) $+\frac{1}{2}$ (measure $\widehat{PC}$) $= \frac{1}{2}$ (measure $\widehat{AC}$).

The proof of Case 2 is left to the student.

Exercise Group 12–11

1. If $\angle ABC$ is inscribed in a circle and measure $\widehat{AC} = 80^\circ$, how many degrees in $\angle B$?

2. An angle of 100° is inscribed in a circle. How many degrees are there in the arc it subtends?

3. Prove that an angle "inscribed in a semicircle" is a right angle (Fig. 12–23).

4. How large can an angle be that is inscribed in a circle?

5. Prove that two inscribed angles in one circle which subtend arcs of equal measure are congruent.

6. Prove that a parallelogram inscribed in a circle must be a rectangle. More generally, prove that if a quadrilateral is inscribed in a circle, then the opposite angles are supplementary.

7. In Fig. 12–24 lines AB and CD are parallel. Prove that measure $\widehat{AC}$ = measure $\widehat{BD}$.

8. A trapezoid is inscribed in a circle. Prove that the base angles are congruent. [*Hint:* Use the result of Exercise 7.]

*9. In Fig. 12–25, AC is tangent to the circle. Prove that measure $\angle ABD = \frac{1}{2}$ measure $\widehat{BD}$. [*Hint:* Draw $DE \parallel AC$. Show that arcs $\widehat{BD}$ and $\widehat{BE}$ have the same measure. Then use $\angle ABD \cong \angle BDE \cong \angle BED$.]

*10. Prove, for Fig. 12–26, that

measure $\angle A = \frac{1}{2}$ (measure $\overset{\frown}{CD}-$ measure $\overset{\frown}{BE}$). State this theorem in words.

*11. State and prove theorems similar to that of Exercise 10 for cases in which either AC or AD, or both, are tangent to the circle.

*12. Prove, for Fig. 12–27, that measure $\angle AEB = \frac{1}{2}$ (measure $\overset{\frown}{AB}+$ measure $\overset{\frown}{DC}$). [*Hint:* Draw BC.] State this theorem in words.

13. Using Fig. 12–27, prove that $\triangle AED \sim \triangle BEC$. Show that $\overline{AE} \cdot \overline{EC} = \overline{BE} \cdot \overline{ED}$. State this last property as a theorem.

*14. Using Fig. 12–26, prove that $\triangle AEC \sim \triangle ABD$. Hence prove that $\overline{AB} \cdot \overline{AC} = \overline{AE} \cdot \overline{AD}$. State this last property as a theorem.

15. Two chords of a circle intersect as in Fig. 12–28. If $\overline{AO} = 12$, $\overline{OB} = 4$, and $\overline{CO} = 8$, find $\overline{OD}$. [*Hint:* Use Exercise 13.]

*16. From point A outside a circle, secants are drawn as in Fig. 12–29. If $\overline{AD} = 6$, $\overline{DE} = 10$, and $\overline{AB} = 8$, find $\overline{BC}$.

*17. In Fig. 12–30, AB is tangent to the circle at B. Prove that $\triangle ACB \sim \triangle ABD$. You will need the result of Exercise 9 for this. Prove also that $\overline{AB}^2 = \overline{AD} \cdot \overline{AC}$.

18. If, in Fig. 12–30, $\overline{AC} = 8$ and $\overline{CD} = 10$, determine $\overline{AB}$.

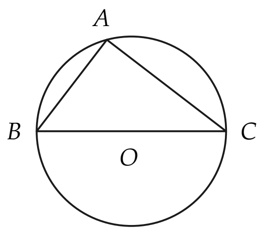

FIGURE 12–23

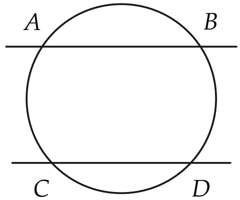

FIGURE 12–24

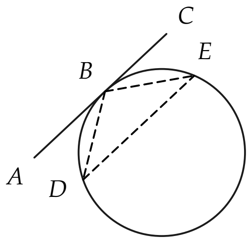

FIGURE 12–25

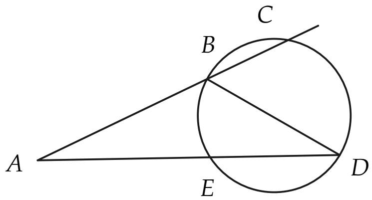

FIGURE 12–26

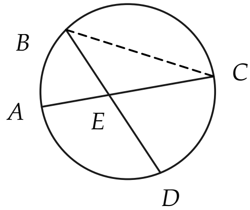

FIGURE 12–27

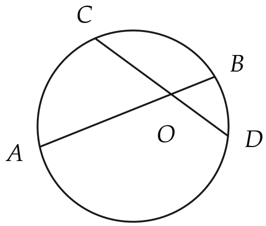

FIGURE 12–28

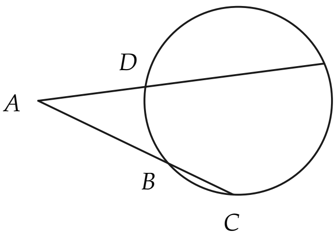

FIGURE 12–29

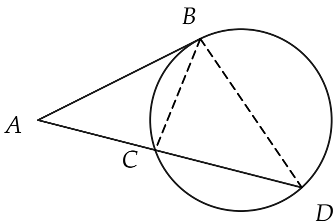

FIGURE 12–30

REVIEW OF CHAPTER 12

In this chapter definitions have been given of the following terms; be sure that you can state them in your own words: arc, subtended arc, inscribed polygonal path, arc length, degree, measure of angle in degrees, measure of arc in degrees, radian, measure of angle in radians, inscribed angle.

The following is an outline of the important definitions and theorems of the chapter. Study the outline so that you understand clearly how each step leads to the next.

Theorem: The limit of the lengths of a sequence of regular inscribed polygons exists.
Definition: Arc length.
Theorem: Can use inscribed polygons other than regular ones to define arc length.
Theorem: Length of "sum" of two arcs equals sum of lengths.
Theorem: Greater angle subtends longer arc.
Theorem: Angles congruent if and only if arc lengths equal.
Theorem: Arc length is proportional to radius.
Theorem: For every positive number less than the circumference, there is an arc having length equal to the number.
Definition: Measure of angles in degrees.
Definition: Measure of angles in radians.
Theorems: Angle measurement and counting.
Theorem: Measure of inscribed angle is one half measure of subtended arc.

Important Remark

The purpose of the chapter has been to define measurement of angles. To make the definition, we first defined *arc length*, and then *angle measurement* in terms of an arc length.

Review Exercises

1. From the definition of arc length, show that the length of arc subtended by a right angle (Fig. 12–31) is greater than $\sqrt{2}\,r$. Hence, conclude that $\pi/2 > \sqrt{2}$.

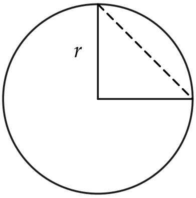

FIGURE 12–31

2. Angles of 15° subtend arcs of lengths 10 and 12 on two circles. What is the ratio of the radii of the circles? Find the radii.

3. Theorem 12–7 states that arc length is proportional to the radius. What theorem about triangles is used in the proof of Theorem 12–7?

4. State briefly what is involved in proving that the length of the "sum" of two adjacent arcs is equal to the sum of their lengths.

5. Compute the length of an arc in a circle of radius 10 if it is subtended by an angle of (a) 40 degrees, (b) 2 radians, (c) $\pi/4$ radians, (d) $\frac{1}{2}$ degree.

*6. A sequence of points A_1, A_2, A_3, $\ldots$ is marked on a circle so that each pair of consecutive points determines a central angle of 1 rad. The direction from A_1 to A_2 to A_3, etc., is counterclockwise. Compute the approximate length of the arc from A_{51} to A_1 in a counterclockwise sense if the radius of the circle is 10. Use $\pi = 3.1416$.